Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/29/2011
PI Name:	Barton, Elisabeth Ph.D.		
Project Title:	Identification of mechano-sensors to protect against skeletal muscle atrophy		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	19104	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	05/01/2009	**End Date:**	04/30/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Addition of Dr. Anastasios Philippou, post-doctoral researcher		
COI Name (Institution):			
Grant/Contract No.:	NNX09AH44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Skeletal muscle mass is maintained by a dynamic equilibrium between signals driving muscle growth and muscle atrophy. Protection against loss of muscle mass in anti-gravity requires enhancement of pro-growth signals to counter the drive toward atrophy. The muscle membrane harbors two protein complexes - the focal adhesion complex and the dystrophin glycoprotein complex – that are important for sensing stress at the membrane, and are thought to help coordinate the balance between growth and atrophy. Whether these complexes mediate identical, complementary, or divergent mechanical signals has been difficult to ascertain, yet both appear to be important for promoting muscle cell survival and muscle growth. Identifying the specific pathways utilized by these mechano-sensors is critical to the development of strategies to maintain muscle mass even in the absence of environmental cues for growth. We will employ an in vitro system to map out the signal transduction pathways triggered by mechanical stress via the focal adhesion and dystrophin glycoprotein complexes, the resultant changes in gene expression, and muscle cell survival. Specific inhibition of each complex will be achieved through isolation of muscle cells from mice lacking members of each complex. These studies will establish a mechanistic understanding of the mechanical responses through each membrane complex, and provide the groundwork for developing novel strategies to prevent muscle atrophy in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Sensing load is a part of daily maintenance of muscle mass and strength, which is affected in anti-gravity, where there is no load on the muscle, but can also be affected in individuals subjected to bed rest, wheelchairs, or casts due to fractures. Understanding how the load sensing mechanisms work can lead to strategies which stimulate the load sensors in the absence of load, and provide a way to maintain muscle mass even when the endogenous signals drive atrophy. A second impact of our specific testing will be in the genetic diseases of muscle, namely the muscular dystrophies. We have confirmed that the sarcoglycan complex is required for generating appropriate load sensing, and so when genetic mutations lead to its loss, the load signals are blunted, and then muscles lose mass. While the most straight froward way to combat these mutations is to replace the mutated gene product, we believe this work will lead to novel treatment strategies for many types of muscular dystrophy.
Task Progress:	The goals of this project are to understand the contribution of a protein complex in muscle cell membranes to sensing load. Using cultured cells from muscle of mice that lack specific members of this protein complex, called the sarcoglycan complex, enables us to test if the complex was important for normal load signaling. First, we grew muscle cells from the mice on flexible membranes so that we could stretch the cells after they matured, and thereby change dynamic load. We found that there were changes in the normal signaling pattern, not only at rest, but also in response to stretch when the sarcoglycan complex was missing. In short, it appeared that the muscles from the normal cells had a transient signaling response in all pathways we measured after a period of stretch. However, the cells lacking the complex had no increased responses in the classical load signaling pathways, but also had heightened responses in the pathways directly coupled to protein synthesis. We conclude that the sarcoglycan complex is a central component of load sensing in muscle. Next we examined changes in gene expression in the two cultures. We found that the absence of the sarcoglycan complex caused increased expression of genes in the “atrogene” pathway, as well as increased expression of IGF-I. Thus, these cultures appear to drive both growth and atrophy pathways when the sarcoglycan complex is absent. At this point, we do not know if these changes result in changes at the protein level, or which pathway ultimately controls the state of the muscle growth. The responses did not change dramatically after a short stretching period, so longer stretching protocol will be used to determine if they evoke more dramatic changes in gene expression. Finally, we were given the opportunity to dissect muscles from mice subjected to anti-gravity and age and sex matched ground controls on the last space shuttle mission, STS-135. These will be used to extend our understanding of how the sarcoglycan complex is modified by lack of load, including localization and post-translational modifications (phosphorylation). We obtained tibialis anterior, extensor digitorum longus, and masseter muscles from these mice. While much has been achieve in previous shuttle missions with mouse samples, the sarcoglycan complex involvement has not been addressed. From related projects, we have reported that there are clear differences between the masseters and TA muscles in terms of loading signals and expression differences. We hypothesize that the loading of masseter muscles comes in part from normal chewing activity, and so the mouse masseters may be spared from atrophy. However, this remains to be tested.
Bibliography Type:	Description: (Last Updated: 09/15/2015)
Articles in Peer-reviewed Journals	Philippou A, Minozzo FC, Spinazzola JM, Smith LR, Lei H, Rassier DE, Barton ER. "Masticatory muscles of mouse do not undergo atrophy in space." FASEB J. 2015 Jul;29(7):2769-79. http://dx.doi.org/10.1096/fj.14-267336 ; PubMed PMID: 25795455 , Jul-2015
Articles in Peer-reviewed Journals	Moorwood C, Philippou A, Spinazzola J, Keyser B, Macarak EJ, Barton ER. "Absence of gamma-sarcoglycan alters the response of p70S6 kinase to mechanical perturbation in murine skeletal muscle." Skelet Muscle. 2014 Jul 1;4:13. eCollection 2014. http://dx.doi.org/10.1186/2044-5040-4-13 ; PubMed PMID: 25024843; PubMed Central PMCID: PMC4095884 , Jul-2014